Course	Computing Algorithms
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Sheet 1

Question 1:

Rank these functions according to their growth, from slowest growing (at the top) to fastest growing (at the bottom).

$$2^n, n^2, n^3, (3/2)^n, n, 1$$

Rank these functions according to their growth, from slowest growing to fastest growing

$$8n^2, 6n^3, 4n, 8^{2n}, n \log_6 n, n\log_2 n, \log_2 n$$

Question 2:

Match each function with an equivalent function, in terms of their O. Only match a function if f(n)=O(g(n))

F(n)	g(n)
$n + 30$	n^4
$n^2 + 2n - 10$	$3n - 1$
$n^3 * 3n$	$n^2 + 3n$
$\text{Log}_2 x$	$\text{Log}_2 2x$

Question 3:

What is the time complexity of the following algorithms and the Big O?

Part 1:

```
int t=0;
for(int i=1; i<=n; i++)
for(int j=0; j*j<4*n; j++)
        for(int k=1; k*k<=9*n; k++)
                t++;
```

Part 2:

```
int x=0;
      for(int i=4*n; i>=1; i--)
           x=x+2*i;
```

Part 3:

```
int z=0;
int x=0;
    for (int i=1; i<=n; i=i*3){
           z = z+5;
           z++;
           x = 2*x;
  }
```

Part 4:

```
int y=0;
    for(int j=1; j*j<=n; j++)
           y++;
```

Part 5:

```
int b=0;
for(int i=n; i>0; i--)
           for(int j=0; j<i; j++)
                  b=b+5;
```

Part 6:

```
int y=1;
int j=0;
   for(j=1; j<=2n; j=j+2)
           y=y+i;
int s=0;
   for(i=1; i<=j; i++)
           s++;
```

Part 7:

```
int b=0;
for(int i=0; i<n; i++)
            for(int j=0; j<i*n; j++)
                  b=b+5;
```

Part 8:

```
int x=0;
for(int i=1; i<=n; i=i*3){
       if(i%2 != 0)
            for(int j=0; j<i; j++)
                   x++;}
```

Part 9:

```
for(int i =0 ; i < =n ; i++)
            for (int j =1; j<= i * i; j++)
                   if (j % i == 0)
                          for (int k = 0; k<j; k++)
                               sum++;
```

Part 10:

```
int a = 0;
int k = n*n;
while(k > 1){
            for (int j=0; j<n*n; j++)
                           { a++; }
                           k = k/2;}
```